Key elements of flow assurance in carbon capture and storage

The challenges and strategies involved in optimising CO_2 flow for safe and effective storage

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Emerging technologies such as carbon capture and storage (CCS) are crucial for reducing and removing anthropogenic carbon dioxide (CO_2) emissions from the atmosphere. Since the Paris Agreement in 2015, countries worldwide have set targets to limit global average temperatures from reaching 1.5°C above pre-industrial levels. The urgency to meet these targets has driven the development of CCS projects from previous concepts of CCS, such as enhanced oil recovery (EOR), which began in the 1970s. There are now 50 commercial-scale CCS projects operational and 534 in development worldwide *(Global CCS Institute, 2024)*.

Flow assurance plays a critical role in the CCS process, ensuring that CO_2 can be transported from its point of capture to its permanent storage site without disruptions. This involves maintaining CO_2 in its supercritical state during pipeline transport, managing pressure and temperature to prevent phase changes, and addressing potential risks like corrosion and blockages. New and developing CCS projects can anticipate and mitigate these challenges by making flow assurance a key factor in the successful deployment of CCS at scale, drawing on decades of experience with CCS in the oil and gas industry.

Selecting the appropriate subsurface storage location is also a crucial component of any CCS project. Using data-driven tools such as bMark can allow screening of potential storage prospects to help identify the optimum storage site. This article explores the intricacies of flow assurance in CCS, highlighting the challenges and strategies involved in optimising CO_2 flow for safe and effective storage.

Supercritical CO_2

CO_2 can be transported in any form, but it is often compressed into a liquid because it occupies significantly less volume compared to its gaseous form. When transported via pipeline, it is most efficient for CO_2 to be in its supercritical state with pressures higher than 74 bar and temperatures higher than 31°C. This represents highest point in which it can exist as a vapour and liquid in equilibrium *(TWI, 2010)*. In this state, CO_2 has

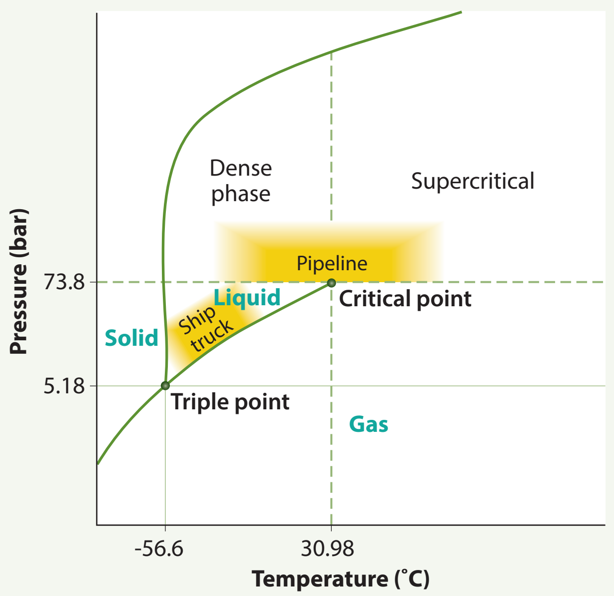

Figure 1 Phase diagram of CO_2, showing the estimated areas of operation for transport via ship, truck, and pipeline
Source: Simonsen et al., 2023

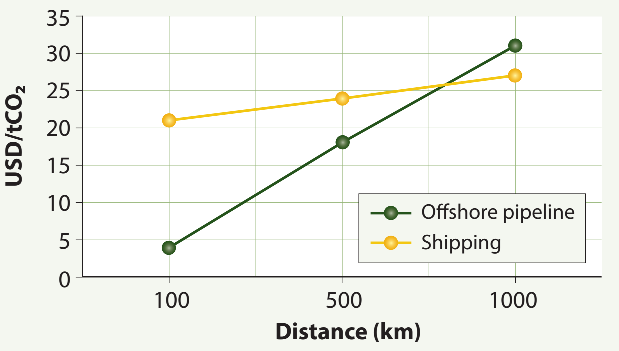

Figure 2 Cost (in USD) of transporting CO_2 via offshore pipelines and ship for distances of 100-1,000 km, assuming a capacity of 2 Mt/year

Source: IEAGHG (2020)

the density of a liquid but the viscosity of a gas *(Drax, 2022)*. The viscosity is up to 100 times lower than that of the liquid phase *(TWI, 2010)*, significantly reducing drag during pipeline flow. This improvement increases CO_2 throughput, leading to lower operational costs.

Figure 1 shows the various transport methods for CO_2 associated with the phase during operation. The difficulty with transporting CO_2 in this state is that it must remain supercritical, requiring the temperature and pressure to be maintained to avoid phase changes.

Pipeline transport of supercritical CO_2

Various methods for transporting CO_2 are available, including pipelines, ships, road, and rail. However, pipelines remain the most cost-effective and widely used option, particularly for long distances and larger volumes of CO_2. As shown in **Figure 2**, pipelines generally have a lower cost compared to ships for distances up to 800 km. Offshore pipelines, especially in regions like the North Sea, will be crucial for the development of commercial-scale CCS projects.

Currently, the US leads the world in the number of operational and developing CCS projects, supported by its extensive 5,000+ mile onshore CO_2 pipeline network. Many existing pipelines previously used for oil and gas can be repurposed for CO_2 usage, which helps to reduce the cost significantly. Reusing pipelines typically costs around 1-10% of constructing new ones *(Drax, 2022)* while also minimising the need for new infrastructure. However, to meet climate targets in the next few decades, approximately 100 times more pipeline infrastructure than currently available will be required *(Global CCS Institute, 2018)*.

To support the expansion of CCS pipeline networks, tools like bMark provide access to up-to-date data sources, including information on existing oil, gas, and CO_2 pipelines worldwide. This data is essential for planning, designing, and scaling up pipeline infrastructure effectively. **Figure 3** highlights the pipeline data for the US, available on bMark.

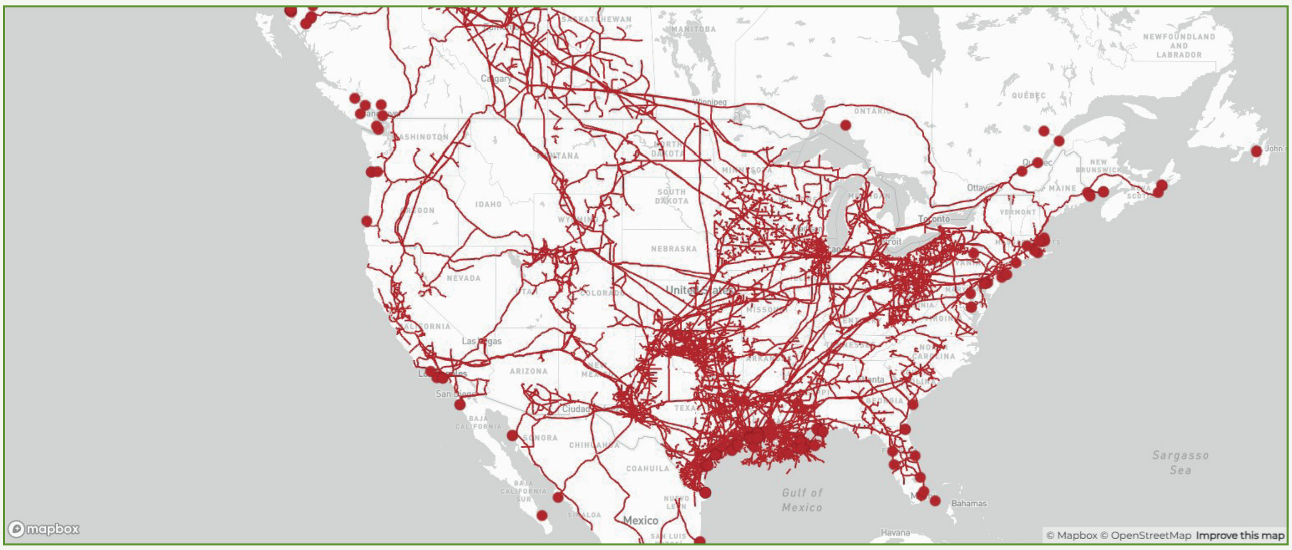

Figure 3 Pipelines data across the US

Source: Belltree bMark

Safety of CO_2 transport

One of the main public concerns about CCS is the safety of CO_2 transport in pipelines and its secure storage underground. However, in EOR projects, CO_2 has been transported over decades, so it is not an unfamiliar technology. It poses no greater risk than oil and gas transport, which has a long history of safe management. Additionally, transporting CO_2 is much safer than other substances as it is not explosive or flammable when mixed with air (*Drax, 2022*). A combination of governmental legislation and international standards will ensure maximum safety in CO_2 transport. Ultimately, one of the most important steps in safeguarding CO_2 transport via pipeline is flow assurance. This critical process involves understanding and mitigating the factors that could disrupt the flow of CO_2 from the capture point to its storage location.

Flow assurance

Flow assurance was initially based on ensuring the successful and economical flow of hydrocarbons from the reservoir to their destination. However, this now directly applies to the transport and injection of CO_2 during CCS projects. Flow assurance studies are an essential part of the design process for oil and gas operations, such as the front-end engineering and design (FEED) process. Factors affecting flow assurance include the formation of solids/blockages, pressure drop, temperature variation, and purity of the CO_2.

Among the factors involved in flow assurance, the primary concern is the potential for blockages within pipelines, whether it is hydrocarbons or CO_2. Studies focus on preventing and controlling the formation of solids which may block the flow. Many things can affect this, especially the pressure, temperature, and chemistry of the product flowing through the pipeline.

Pressure drop is one of the key concerns in flow assurance, with many contributing factors affecting wellbore pressure, fluid properties, and temperature. Here again, CO_2 injection projects can use the experience gained from the oil and gas industry. In hydrocarbon production, wellbore pressure is a significant factor in pressure drop, with up to 80% of the lost pressure occurring during the flow from the subsurface to the surface.

Fluid properties, such as viscosity and density, are also crucial. Fluids with lower density flow with less pressure drop, while higher viscosities cause more friction, leading to a greater pressure drop (*Böser and Belfroid, 2013*). Temperature also influences flow assurance; higher temperatures typically decrease viscosity and density, which can limit pressure drop. However, temperature variations can also lead to a drop in pressure.

While these factors affecting hydrocarbon flow are present in terms of the flow of CO_2, the effects vary. The pressure drop due to the wellbore will not be the same with CO_2 because it is injected rather than produced. The fluid properties of CO_2 differ from hydrocarbons, and CO_2 is usually transported in its dense or supercritical phase. The purity of CO_2 and the presence of impurities can affect its viscosity and density in pipelines, nitrogen (N_2) having the greatest and hydrogen sulphide (H_2S) having the least impact on density and pressure loss.

“Among the factors involved in flow assurance, the primary concern is the potential for blockages within pipelines, whether it is hydrocarbons or CO_2”

The presence of certain impurities, including water (H_2O), sulphur oxides (SOx), and nitrogen oxides (NOx), can also affect the flow of CO_2 by increasing the risk of corrosion. Any water content within the pipeline can lead to corrosion, so an industry-standard limit of 500 ppm is maintained to minimise the risk (*Simonsen, et al., 2024*). Conversely, the presence of other impurities, such as N_2, can bring down the water concentration levels in CO_2. In supercritical CO_2 environments, the addition of O_2 can increase corrosion in carbon steel, with severe corrosion occurring when both O_2 and H_2S are present. Therefore, the capture and processing of CO_2 for storage projects is critically important. Furthermore, some pipelines may need to be designed with specific materials to account for the presence of certain impurities, to reduce the risk of corrosion.

Single and two-phase flow

The transport of CO_2 is currently limited to single-phase flow, either in its dense or supercritical form. Existing tools for the flow assurance of hydrocarbons can be used for CO_2 if it remains in a single phase. If it is kept at high pressure and temperature, CO_2 will stay in its single, supercritical phase. However, as discussed previously, any leaks or depressurisation will risk CO_2 going into two phases. The two-phase flow of CO_2 is when the gas and liquid forms coexist. The issue with two phases in flow assurance is the potential for severe slugging within the pipeline, which involves the intermittent flow of gas and liquid.

Slugging can disrupt the flow, causing significant pressure and temperature variations that may cool pipelines, making them brittle and prone to rupturing (*Yang et al., 2021*). Again, these issues have been learned from the oil and gas industry, where slugging is a major challenge during production. Due to these disruptions in flow, various methods have been developed to identify and handle changes in phase, including dynamic models based on mass, momentum, and energy conservation. To manage the potential of phase changes within CO_2 pipelines, it is crucial to control the temperature and pressure within the pipeline to ensure stable conditions.

Reservoir injection

As the flow of CO_2 reaches its endpoint at the storage site in the subsurface, flow assurance remains equally important due to the change in pressure. The high pressures required to ensure CO_2 remains in its supercritical state could be met by the lower pressures at the injection point, depending on the site. For example, injection into a depleted hydrocarbon field with low pressures can result in adiabatic cooling, with the reduced temperatures possibly leading to hydrate formation and freezing of pore water (*Loeve et al., 2014*).

CO_2 injection into an aquifer is less likely to have the risk of pressure change and freezing, as no fluids have been extracted, therefore pressure remains stable. However, the risk of lower pressure within aquifers remains, depending on factors such as the depth of the aquifer, the height of the water column, and nearby hydrocarbon production, potentially changing the regional pressure. The advanced capabilities of software like bMark include accessing historical production data from reservoirs, as well as temperature and pressure data (see **Figure 4**), which can help further understand the characteristics of a potential CO_2 storage site.

CCS projects

The selection of appropriate locations for permanent CO_2 storage is also fundamental to

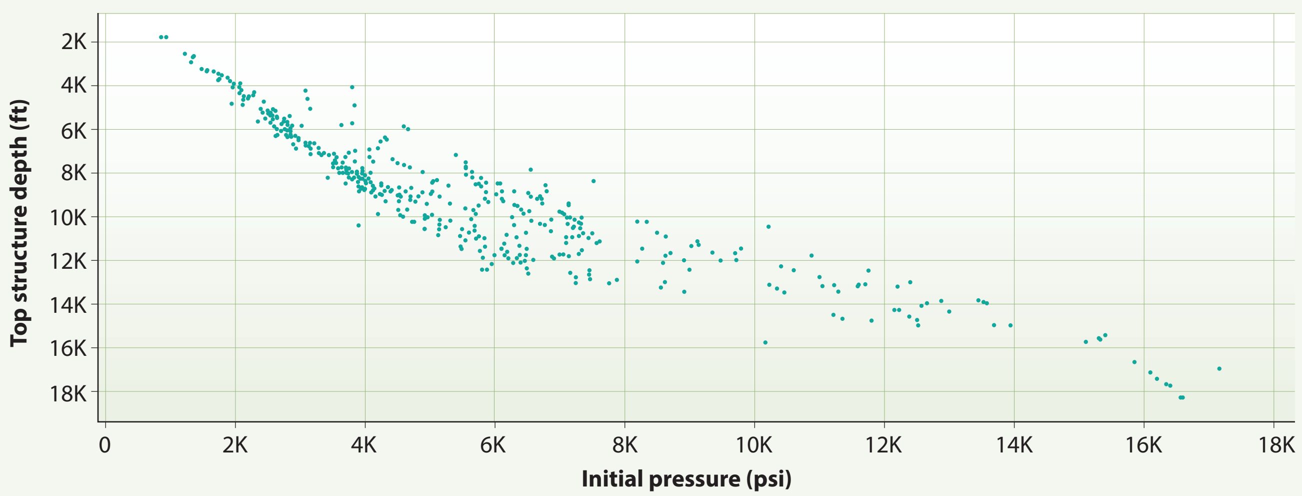

Figure 4 Depth vs pressure plot of fields in the North Sea *Source: Belltree bMark*

Figure 5 CCS projects in Europe *Source: Belltree bMark*

the success of a CCS project. While emissions from numerous industries contribute to global CO_2 levels, the current focus in CCS deployment should be on minimising costs and maximising captured CO_2 by prioritising industrial hubs and clusters.

Also, choosing project sites near existing CO_2 storage locations and infrastructure, such as pipelines, can further reduce costs by minimising transport distances and repurposing existing infrastructure. Tools like bMark are valuable for visualising active and planned CCS projects, identifying nearby pipelines for use, and offering key data such as the geological characteristics of potential storage sites. **Figure 5** highlights the CCS projects that are active or planned in Europe (from bMark).

Summary

As carbon capture and storage emerges as a vital strategy in combating climate change, ensuring the safe and efficient transport and storage of CO_2 becomes paramount. Flow assurance is a critical aspect of CCS, involving the management of factors such as pressure drops, temperature variations, and the risk of corrosion or blockages in pipelines.

Drawing on decades of experience in the oil and gas industry, flow assurance principles used for hydrocarbons are applied to CO_2 to prevent operational disruptions, especially during its transport in supercritical form. The transition from capturing CO_2 to its final geological storage demands meticulous planning, especially as the CO_2 enters subsurface reservoirs where pressure and temperature changes can lead to challenges like hydrate formation or phase changes.

By leveraging data-driven tools and incorporating detailed analyses during the early design stages, CCS projects can mitigate risks and enhance the stability of CO_2 flow throughout the process. As the global push for CCS accelerates, mastering flow assurance will be crucial for the safe and long-term sequestration of CO_2, ultimately contributing to global efforts to limit temperature rise and achieve climate goals.

> **“By leveraging data-driven tools and incorporating detailed analyses during the early design stages, CCS projects can mitigate risks and enhance the stability of CO_2 flow throughout the process”**

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